

What's on the Menu? A Review of School Meal Quality Across 29 Countries

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Abstract

Objective: The double burden of malnutrition (DBM) – the coexistence of undernutrition and overweight/obesity – poses a critical global health challenge, particularly for children and adolescents. School meal programs offer an opportunity to address the DBM by providing nutritious meals that support growth, development, and lifelong health. However, limited school meal quality data hinders effective program design. This study evaluates global school meal quality through nutrient composition analyses and the Global Diet Quality Score (GDQS)-Meal and -Menu metrics.

Design: Data were collected from the Global Child Nutrition Foundation's 2024 Global Survey of School Meal Programs, grey literature, and in-country stakeholders. Nutritional content was compared to age-specific Nutrient Reference Values, including Harmonized Average Requirements. The nutritional quality of meals and menus was assessed using GDQS metrics based on 25 food groups.

Setting: Twenty-nine countries across diverse geographic and socioeconomic contexts.

Participants: Not applicable.

Results: Most menus met or exceeded 30% of recommended daily nutrient values. Lunches had the highest nutrient contributions, followed by snacks and breakfasts. GDQS-Meal and -Menu scores revealed variability across age groups, meal types, and countries. Overall, 57% of menus achieved high nutritional quality, 37% medium, and 6% low. Few menus lost points for unhealthy components. Diverse, balanced menus with healthy food groups scored higher, though fortification also proved nutritionally valuable.

Conclusions: While most menus contribute significantly to daily nutrient needs, variability across countries, age groups, and meal types highlights opportunities for context-specific improvements through diversification and fortification. Learning from high-performing peer programs can help identify feasible improvements.

Keywords: School Meals, Nutrition, Children, Double Burden, Menu Quality

Introduction

Double Burden of Malnutrition in Children

The double burden of malnutrition (DBM), characterized by the coexistence of undernutrition (stunting, wasting, and micronutrient deficiencies) alongside overweight and obesity, represents a critical global health challenge.^(1–5) This dual burden is a leading risk factor for diet-related non-communicable diseases (NCDs) and contributes significantly to global morbidity and mortality.^(1,6–7) The rise in the DBM is closely linked to the ongoing nutrition transition, where dietary patterns shift from traditional diets to processed, calorie-dense, nutrient-poor foods.⁽⁸⁾ Urbanization, sedentary lifestyles, reduced physical activity, and limited access to affordable, nutritious food exacerbate this complex landscape.⁽⁸⁾

Although the DBM affects individuals across life stages, children and adolescents are particularly vulnerable.⁽⁴⁾ Childhood and adolescence are critical phases of growth and development, marked by heightened macro- and micronutrient needs to support rapid physical and mental growth.⁽⁹⁾ Inadequate nutrition during these periods can lead to lifelong health risks.⁽⁹⁾ Malnutrition, encompassing both overweight/obesity and undernutrition, in children is associated with increased susceptibility to communicable and non-communicable diseases, weakened immune function, delayed motor development, impaired cognition, compromised reproductive health, and elevated mortality rates.^(7,10–11) Undernutrition manifests as stunting (low height-for-age), underweight (low weight-for-age), and wasting (low weight-for-height), typically caused by insufficient protein-energy intake.^(7,12) Micronutrient deficiencies, which result from inadequate intake of essential vitamins and minerals, affect over half of preschool-aged children worldwide.⁽⁷⁾

While the prevalence of undernutrition has declined globally over the last 30 years, particularly in low- and middle-income countries, the rates of childhood overweight (Body Mass Index [BMI]-for-age greater than 1 standard deviation above the WHO Growth Reference median) and obesity (BMI-for-age greater than 2 standard deviations above the WHO median) have continued to rise.^(4,12) From 1990 to 2022, the prevalence of overweight among children and adolescents surged from 8% to 20%, with over 390 million children and adolescents overweight in 2022.⁽¹²⁾ Overweight and obesity primarily result from an imbalance between energy intake and expenditure, influenced by obesogenic environments, psychosocial factors, genetic predispositions, and structural barriers, such as limited access to

healthy, affordable foods and opportunities for physical activity.⁽¹²⁾ These conditions often persist into adulthood, significantly increasing the risk of developing NCDs, including cardiovascular diseases, diabetes, chronic respiratory diseases, digestive disorders, neurological disorders, and certain cancers.^(12–14) Additionally, childhood obesity increases the risk of developing psychosocial and emotional problems and disorders, which adversely influence school performance and quality of life.^(12,14)

Role of School Meal Programs

These adverse effects associated with the DBM highlight the need for comprehensive, scalable programs that support adequate nutrient intake during childhood. Children spend a significant portion of their waking hours at school, and school meal programs, which can provide 20%-70% of children's daily energy needs, are uniquely positioned to shape dietary habits and preferences.^(15–17) School meal programs have been recognized as essential safety nets with documented impacts across social protection, educational, health, and nutritional dimensions.^(18–19) These programs can serve as double-duty actions (DDAs) for addressing the DBM.^(2,9,19) DDAs, grounded in the notion that under- and overweight/obesity share common drivers, encompass cross-sector actions that target all forms of malnutrition simultaneously.⁽¹⁾ Reaching at least 408 million children in 148 countries, school meal programs can be leveraged as meaningful opportunities to improve dietary quality, enhance child nutrition, and reduce the global DBM.^(15–16,20–22)

Gaps in School Meal Quality Data

The effectiveness of school meal programs depends on the quality of their food baskets (the collection of food items and ingredients included in planned menus provided to school children).⁽²³⁾ School meal programs are incredibly complex and diverse, differing across and within countries.^(16,24) Programs vary by scope (national, regional, local), geographic setting (urban, suburban, rural), modality (in-school meals, in-school snacks, take-home rations), meal frequency and type (breakfast, lunch, snack, dinner), procurement sources (local farms, school gardens, community provision, private suppliers), inclusion of fortified or biofortified foods, and costs.^(5,16,25) Therein, school food basket contents are also diverse. School food baskets are typically set by menus – a defined list of one or more dishes planned as a vehicle to ensure that school meals meet their proposed aims.⁽²⁶⁾

However, data on school menu quality remains limited, hindering efforts to design and implement effective programs tailored to diverse populations and nutritional needs. Existing studies primarily focus on school feeding coverage or qualitative aspects of meal quality, such as assessments of frequency or variety, with few rigorously evaluating meal or menu quality.^(26–27) This study provides a cross-country review of the quality of school meal program menus, focusing on nutrient composition and menu quality. Specifically, this research addresses the following questions:

- (1) How does the nutritional quality of school meals and menus compare across countries?
- (2) How does school meal and menu quality vary by age group (preschool, primary, secondary) and meal type (breakfast, lunch, snack)?
- (3) What factors contribute to variations in school meal and menu quality?

To our knowledge, this is the first study to jointly apply nutrient composition analysis and food-based quality assessment to a cross-country school menu dataset, offering a multidimensional lens for evaluating school meal quality and addressing the DBM. It is also among the first to apply GDQS-Meal and -Menu metrics to real-world cases, and the first to apply these metrics to school menus across a wide range of country income levels.

Methods

Data Collection

Data on school meals were collected from the Global Child Nutrition Foundation's (GCNF) 2024 Global Survey of School Meal Programs (GSSMP), supplemented by a grey literature search drawing from in-country stakeholders and program websites to identify additional menus. The grey literature search followed a structured protocol involving multilingual Google searches using country and program names and exploration of relevant ministry websites. Launched in 2019 and repeated in 2021 and 2024, the GSSMP collects detailed information on national- or large-scale school meal programs.^(18,25,28) The 2024 GSSMP also gathered sample menus or guidelines, providing a direct look at the food baskets planned for school children. The 2024 survey received a response from 142 country governments, with 125 reporting large-scale school feeding activities and sharing detailed information on 207 individual school meal programs.⁽²¹⁾ A total of 21 quantitative program menus with adequate menu data were identified from the survey responses, and an additional 10 menus were identified through the grey literature search.

An extraction sheet was developed to systematically analyze survey responses. Responses were reviewed to identify age groups served (preschool, primary, secondary); menu type (menu – a set of meals served over a specified time-period, meal – breakdown of a single eating occasion which may include multiple dishes and foods, or food – an unstructured list of food items served); meal types (breakfast, lunch, snack); data type (quantitative/qualitative); and the number of days of information. Age group classifications and meal types varied from country to country, but generally, preschool corresponded to approximately 3-5 years, primary to 5-9 years, and secondary to 10 years or older.⁽²⁵⁾ Country focal points were contacted for missing information or additional clarification as needed. Menus not in English, German, or French were translated using Google Translate, and program-level quantitative menu and recipe data were extracted.

Data Analysis

Food basket items were matched to the United States Department of Agriculture's (USDA) food composition table (FCT) (SR Legacy) using Excel, which retrieved the relevant USDA food codes and nutritional values.⁽³¹⁾ FCTs provide the nutrient composition of commonly available and consumed foods.^(19,31) USDA's FCT is among the most widely used and internationally recognized nutrient databases, offering standardized and comprehensive nutrient profiles of a wide variety of foods.⁽³²⁾ While region-specific FCTs are often ideal, they do not always offer complete or standardized data.⁽³²⁾ Additionally, cross-country comparisons are prone to error when different FCTs are used to estimate nutrient intake.⁽³²⁾ The matched FCT-food basket items were analyzed using statistical software R (version 4.3.1).

Nutritional content was compared to a combination of Nutrient Reference Values (NRVs). Primary comparisons used Harmonized Average Requirements (H-ARs) – harmonized NRVs by nutrient, age group, and sex designed for global intake assessments across populations.⁽²⁹⁾ However, H-ARs do not include energy or macronutrient references, so energy content was assessed using FAO/WHO/UNU Expert Consultation Human Energy Requirements, and macronutrients compared to the Dietary Reference Intakes and WHO guidelines, which offer age- and sex-specific guidance widely used in both research and policy.^(29–30,33–34) One challenge in establishing universal nutrient requirements for school meals is the variation in age groups served, making it difficult to apply a single cut-off across all students. Although

there is no universally-accepted standard for school meal nutrient contributions, most national and global school meal frameworks recommend that meals contribute at least 30% of children's daily nutrient intake.^(19,35–38)

The Global Diet Quality Score (GDQS)-Meal and -Menu metrics were applied to assess meal and menu quality based on the intake of 25 food groups (16 healthy food groups, 7 unhealthy food groups, and 2 food groups that are healthy when consumed in moderation but unhealthy when consumed in excess); the presence of fortification and/or biofortification; and the diversity of healthy food groups.⁽³⁹⁾ These food-based metrics, designed for global school meal program applications, offer comprehensive standardized assessments of meal and menu quality across different countries and regions.⁽³⁹⁾ By evaluating both healthy and unhealthy components, this dual perspective reflects both sides of the DBM.⁽³⁹⁾

The GDQS-Meal awards points for the presence of healthy foods (*GDQS-Meal+*), the absence of unhealthy foods (*GDQS-Meal-*), the inclusion of fortified and biofortified foods considering 16 different micronutrients (*GDQS-Meal Fortification/Biofortification*), and the diversity of healthy food groups (*GDQS-Meal Food Group Diversity*). The total score is calculated by adding all the sub-metrics (*GDQS-Meal+*, *GDQS-Meal-*, *GDQS-Meal Fortification/Biofortification*, and *GDQS-Meal Food Group Diversity*) for a maximum raw score of 83 and a maximum scaled score of 100. All raw scores are translated into a scaled score, which is used for reporting and benchmarking purposes.⁽³⁹⁾ The GDQS-Menu extends this assessment to multiple meals over a defined period (2-7 days, defined by user).⁽³⁹⁾ It calculates the average *GDQS-Meal+* and *GDQS-Meal-* scores across the menu period, and adds points based on the total number of distinct fortificants/biofortificants and healthy food groups included across days (*GDQS-Menu+*, *GDQS-Menu-*, *GDQS-Menu Fortification/Biofortification*, and *GDQS-Menu Food Group Diversity*).

Both GDQS-Meal and -Menu metrics are age-specific, with gram-based food group thresholds for three age groups: 24-59 months, 5-9 years, and 10+ years. Meals and menus are not expected to achieve a full 100 points, as the scoring is not designed to prescribe ideal meals or menus. Rather, it is designed to evaluate the healthiness of diverse meals and menus, achievable through various combinations of foods, food groups, and fortificants/biofortificants considering cultural preferences and contextual factors. Nutritional

quality is interpreted using GDQS-specific benchmarks: scores below 22 reflect low nutritional quality, scores between 22 and 35 reflect medium quality, and scores of 35 or higher indicate high nutritional quality.⁽⁴⁰⁾

Since some programs serve multiple meals per day (breakfast, lunch, snacks), modified versions of the GDQS-Meal and -Menu metrics were developed to generate a more comprehensive daily nutritional profile. These Modified GDQS-Meal and -Menu metrics were calculated by including all meals and food items served on a given day across all meal types to create aggregate daily scores by program. All GDQS-Meal and -Menu metrics, including the Modified GDQS-Meal and -Menu metrics, were calculated using the GDQS-Meal and -Menu Metric Data Collection and Tabulation Excel Tool, which is designed to ensure easy and efficient collection and tabulation of the metrics.⁽⁴¹⁾

Several assumptions were made when matching food baskets to the FCT and GDQS food lists. For programs with multiple weeks of data, it was assumed each week was nutritionally comparable, so the first week was used for analysis. Additionally, it was assumed that fortified foods, when listed, would be fully implemented, regardless of the scale of implementation. For ambiguous food item matches, the most common variant was selected based on context. Approximate equivalents were used when items had no GDQS or USDA FCT match. When not specified, foods were assumed cooked unless typically served raw (i.e., fruits, salads).

Results

Descriptive Statistics

In total, 31 program menus were included in the analysis across 29 countries. Countries with more than one program had distinctions based on age group or meal type; for instance, one country operated separate programs for preschool and primary students, while another had distinct programs for breakfast and lunch. Although some countries, including India, Canada, and the United States, have decentralized school meal programs with state- or province-level variations, only identified subnational program menus were included. Descriptive statistics for each country are presented in Table 1. Additional information on program universality and targeting criteria is provided in Appendix A. School meals and menus were evaluated

across three age groups (preschool, primary, and secondary) and three meal types (breakfast, lunch, and snack).

The majority of countries represented were from Europe & Central Asia ($n = 6$), followed by Latin America & Caribbean ($n = 5$), Sub-Saharan Africa ($n = 5$), South Asia ($n = 5$), East Asia & Pacific ($n = 3$), Middle East & North Africa ($n = 3$), and North America ($n = 2$). The income distribution of countries (based on the World Bank classification) included three low-income countries (LICs), twelve lower-middle-income countries (LMICs), seven upper-middle-income countries (UMICs), and seven high-income countries (HICs).

Primary-age meals were the most common (found in 100% of programs), followed by preschool (45%) and secondary-age meals (41%). Most countries (79%) offered lunch, while breakfast (28%) and snacks (21%) were less common. A majority of countries (76%) had a five-day menu, four reported six-day menus, two reported four-day menus, and one provided a seven-day menu. Two countries provided one-day menus but indicated that meals were provided five days a week; in these cases, the one-day menu was extrapolated to represent a typical five-day offering.

Overall Quality of School Meals

Nutrient Composition Analysis

Table 2 presents the mean daily nutritional composition of school meals across the 29 countries, averaged across age groups, alongside the NRV percentage achieved for each nutrient. Appendix B provides a complete table of the total daily nutritional composition of school meals disaggregated by day, age, and meal type. Figure 1 presents boxplots of the distribution of NRV percentages for each nutrient overall and disaggregated by age group and meal type.

Across all meals and age groups, most nutrients met or exceeded 30% of the NRVs, aligning with global standards recommending meals provide at least one-third of daily nutritional needs.^(19,35–38) Certain micronutrients were consistently provided in abundance: folate had the highest average contribution (124%), followed by vitamin B12 (85%) and vitamin C (82%). Protein (61%) and total sugar (43%) also exceeded 30% of daily recommended values. Energy (25%), saturated fat (24%), fiber (23%), total fat (22%), and sodium (22%) achieved

the lowest NRV contributions. Calcium was also comparatively low, with a mean NRV of 35%.

By age group, preschool-aged children received the most nutrient-dense meals relative to their needs, averaging 70% of NRVs, followed by primary school children (62%), with secondary school children faring less well (43%). Among preschool meals, the highest contributions came from folate (182%), vitamin C (111%), and vitamin B12 (110%), while fiber was the lowest at 26%. Primary school meals also had strong contributions from folate (118%) and vitamin B12 (107%). In contrast, secondary school meals had lower contributions overall, with vitamin C at 63%, calcium at 29%, and iron at 38%. Secondary school lunches and snacks were less frequently fortified, resulting in lower nutrient density relative to needs, particularly for micronutrients. Concurrently, secondary school meals had lower total sugar (30%) and saturated fat (22%) contributions than primary and preschool meals (39% and 41% for sugars; 30% and 41% for saturated fat, respectively).

Lunches (n=45) were the most nutrient-rich meals across all groups, averaging 63% of NRVs. They provided particularly high levels of vitamin B12 (107%), vitamin A (105%), folate (105%), and vitamin C (100%). Snacks (n=12) followed, with an average contribution of 55% across nutrients. Despite offering lower macronutrient content (20% energy, 36% protein, and 7.0% fiber), they often exceeded 100% of the NRV for some micronutrients like folate, especially in preschool (334%) and primary (164%) age groups. This pattern reflects the smaller portion sizes but higher micronutrient density in snacks like fortified biscuits. Breakfasts (n=14) had the lowest overall nutrient contribution at 53%, though they performed relatively well in micronutrients such as thiamin, folate, and riboflavin. Snacks and breakfasts also showed relatively higher NRV percentages for total sugars (46% and 52%, respectively), compared with lunches (31%).

When disaggregated by age and meal type, preschool lunches delivered the highest nutrient contribution (77% of NRVs), followed by preschool snacks (65%). In contrast, secondary school snacks were the least nutrient-dense, contributing 31% on average, followed by secondary school lunches (43%).

GDQS-Meal and -Menu Analysis

Figure 2 illustrates variability in GDQS-Meal and -Menu scores by country, disaggregated by age and meal type, with validated nutritional quality benchmarks.⁽³⁹⁾ GDQS-Meal and -Menu scores varied widely across age groups, meal types, meal days, and countries. Among preschool students, the lowest GDQS-Menu score was recorded in Djibouti (13) and the highest in Uruguay (49). Primary-age GDQS-Menu scores ranged from 13 (Djibouti) to 60 (United States). Secondary-age menus followed a similar pattern, with Djibouti recording the lowest snack score (13) and the United States recording the highest lunch score (60), driven by a high food group diversity score (16). Across all age groups, snacks consistently had the lowest GDQS-Menu scores.

Across all age groups and meal types, most menus (57%) achieved high nutritional quality for their GDQS-Menu scores, while 37% fell into the medium category, and 6% were classified as low. When disaggregated by age, preschool menus (n=15) were evenly split between high (47%) and medium (47%) nutritional quality, with 7% falling into the low category. Among primary-age menus (n=37), 57% were high quality, 38% medium, and 5% low. Secondary-age menus (n=16) had the highest proportion (69%) of high nutritional quality scores, with 25% classified as medium, and 6% as low. The higher proportion of high-quality secondary-age menus may reflect that 75% of these meals were lunches, which generally received higher GDQS-Menu scores, compared to 67% of preschool meals and 65% of primary meals.

Across all age groups, lunches generally scored higher than breakfasts and snacks, typically driven by higher food group diversity. Although lunches had the highest proportion of meals rated as high nutritional quality (70% high, 30% medium), they also exhibited the widest GDQS-Menu score range (27 in Djibouti to 60 in the United States). Lunch compositions varied considerably across countries; for example, Uruguay offered multiple dishes within a single lunch, such as salad, a main dish like lentil-meatloaf, bread, and fruit, while Djibouti provided a single main dish composed of rice or fortified flour, pulses, vegetable oil, and iodized salt.

Breakfasts had a narrower GDQS-Menu score range (22 in South Africa to 44 in Belize), with 64% of breakfasts achieving high nutritional quality, 27% medium, and 9% low.

Breakfast composition also varied, with South Africa offering alternating single-component meals such as instant flavored maize meal or sorghum porridge, while Belize provided breakfasts with multiple components such as egg breakfast wraps and chicken sandwiches. Snacks had the lowest GDQS-Menu scores overall, ranging from 13 in Djibouti to 34 in Bangladesh, and none achieved high nutritional quality. Most snacks (73%) achieved medium nutritional quality, while 27% had low nutritional quality.

Table 3 presents Modified GDQS-Menu scores, averaged across age groups. Appendix C provides a complete table with GDQS-Meal and -Menu scores disaggregated by day, age, and meal type. Modified GDQS-Menu scores ranged from 23 for both Afghanistan and Yemen to 60 for the United States. Countries with higher scores typically served multiple meals per day and multiple dishes within each meal, contributing to higher food group diversity sub-metrics. Additionally, higher-scoring countries generally exhibited more balanced sub-metric distributions, whereas lower-scoring countries earned points primarily from one or two sub-metrics.

Factors Influencing Meal and Menu Quality

Modified GDQS-Menu scores were generally associated with economic status, with higher-income countries tending to achieve higher scores (Figure 3). However, considerable score variability within income groups revealed that GDP per capita is not a definitive predictor of meal quality. Figure 4 presents a breakdown of Modified GDQS-Meal and -Menu sub-metrics by income group (Panel A), country (Panel B), and day (Panel C).

HICs generally exhibited more balanced distributions across GDQS-Menu sub-metrics, with the largest portion of the score (36%) coming from food group diversity, followed by the absence of unhealthy foods (30%), the presence of healthy foods (28%), and fortification (7%). UMICs followed a similar pattern, with 33% of the score from food group diversity, 33% from the absence of unhealthy foods, 25% from healthy foods, and 9% from fortification. In contrast, LICs and LMICs derived most of their points from the absence of unhealthy foods (59% for LICs, 43% for LMICs), with smaller contributions from food group diversity (4% for LICs, 18% for LMICs). LMICs showed a modest contribution from the presence of healthy foods (23%) and limited use of fortification (16%), whereas fortification played a more substantial role in LICs (28%). LMIC and LIC countries such as Afghanistan,

Yemen, and Bangladesh, which rely on fortified biscuits as school meal snacks, recorded the lowest food group diversity scores (0%). Across all income groups, the absence of unhealthy foods consistently encompassed a large portion of the score. Additionally, while broader patterns exist, variability can also be seen within income groups (Panel B) and within countries across different days (Panel C).

Table 4 presents the mean daily intake (in grams) of GDQS food group across all age groups globally and at the country-level. The most consumed food group globally was *Refined Grains and Baked Goods* (98 g/day), classified as unhealthy, followed by *Low Fat Dairy* (84 g/day) and *Other Fruits* (59 g/day), both considered healthy. In contrast, several nutrient-dense healthy food groups were minimally consumed, including *Nuts and Seeds* (2 g/day), *Liquid Oils* (7 g/day), and *Dark Green Leafy Vegetables* (10 g/day). Additionally, 68 g/day of food intake fell into the *Not Classified* category (i.e., spices, butter, water), which, while important for taste, palatability, and hydration, do not contribute substantially to nutrient diversity.

Food group consumption varied widely across countries. Meals in Canada, Costa Rica, and the United States featured high intakes of dairy, especially low-fat varieties. In contrast, India, Sri Lanka, Lesotho, and Djibouti had the highest intakes of *Refined Grains and Baked Goods*. Countries offering greater food group diversity, particularly within healthy food groups, tended to achieve higher Modified GDQS-Menu scores (Table 3). For instance, school meals in the United States, Moldova, Belize, and Uruguay included a broad range of food groups and a high representation of healthy items. In contrast, countries with lower scores reported limited food group variety and relied on fortified products to meet nutritional needs; in Afghanistan, Bangladesh, and Yemen, meals primarily consisted of fortified biscuits while meals in other countries, like Djibouti, Honduras, and South Sudan, were generally based on staples such as cereals or rice, pulses, and vegetable oils.

Discussion

This study provides a cross-country assessment of school meal and menu quality as planned. While existing research on school meal programs has focused on coverage, this analysis makes a unique contribution by systematically evaluating the nutritional quality of meals through a complementary approach combining nutrient composition analysis with GDQS-

Meal and -Menu metrics. This dual emphasis on nutrient- and food-based quality, as well as the inclusion of healthy and unhealthy food components, offers actionable insights for school meal program design and addressing the multiple dimensions of the DBM.

Key Takeaways

1. School Meals are a Critical Source of Nutrition for Children.

On average, school meals provided a substantial share of daily nutrient needs. Across age groups and meal types, most nutrients met or exceeded 30% of NRVs. Certain micronutrients, such as folate (124%), vitamin B12 (85%), and vitamin C (82%), had high contributions to daily needs, underscoring the role school meals can play in preventing micronutrient deficiencies. Concurrently, the analysis highlights opportunities for strengthening nutrient delivery. For example, fiber (23%) and calcium (35%) were less consistently provided.

GDQS-Menu scores affirmed overall quality: 57% of menus achieved high nutritional quality, 37% had medium nutritional quality, and only 6% had low nutritional quality. Higher GDQS-Meal and -Menu scores reflect a diverse, balanced, and nutrient-rich combination of foods.⁽⁴²⁾ Importantly, few menus lost points for unhealthy components, highlighting a shared objective to design school meal programs in pursuit of nutrition goals. This finding reinforces the potential for school meals to not only address micronutrient deficiencies but also reduce the rising burden of diet-related NCDs.

Parallel NRV and GDQS analyses reveal paths for improvement. Fiber intake could be enhanced by replacing some food items in the *Refined Grains and Baked Goods* food group, the most consumed food group globally, with alternatives from the *Whole Grains*, *Dark Green Leafy Vegetables*, or *Nuts and Seeds* groups. These alternative high-fiber foods are also rich in folate, B vitamins, iron, and calcium.⁽⁴³⁾ Budget-neutral substitutions, like those promoted by the Fortified Whole Grains Alliance, may support scalable, cost-effective, and nutritious improvements.⁽⁴³⁾ Rwanda's Fortified Whole Grain Program, for example, produced fortified whole grain maize at the same cost as refined maize, with high student acceptance.⁽⁴³⁾

2. Substantial Variability in Meal and Menu Quality Exists

School meal quality varied across age groups, meal types, and countries. Preschool-aged children received the most nutrient-dense meals relative to their needs (70% of NRVs), followed by primary (62%) and secondary students (43%). GDQS-Menu scores showed less variation: 69% of secondary menus were high quality, compared to 57% for primary and 47% for preschool. This discrepancy may reflect differences in fortification and meal types by age group: preschool and primary children more often received fortified snacks or breakfasts, while secondary students typically received lunches with more food diversity but less fortification. Calcium was consistently low across age groups, suggesting a need for more dairy or other calcium-rich foods. In many contexts, fortified milk could provide calcium and additional micronutrients like iron.⁽⁴⁴⁾ Lower nutrient adequacy in secondary meals also reflects the challenge of meeting adolescents' higher dietary needs when standard meals are served across all age groups, which can result in under-provision for older students. These findings reinforce the importance of aligning meal design with age-specific needs and addressing identified gaps, including iron for adolescents (particularly adolescent girls) to support healthy growth and reduce long-term health risks.⁽³⁸⁾

Within-country variation may reflect budget constraints, staggered nutrient delivery across the week, or efforts to tailor meals to children's preferences. This variability is not inherently negative; it presents opportunities to replicate well-performing meal structures and strengthen weaker components. These differences may also map onto cost variations: more nutrient-rich, diverse menus, particularly those incorporating fresh components, may be costlier to procure and prepare, whereas meals based on fortified or shelf-stable products may be more affordable but offer less variety.⁽²⁴⁾

3. There are Multiple Paths to Achieving Quality

The diversity among high-quality school meals illustrates the multitude of context-specific pathways to achieving quality. While some countries achieved high scores through diverse food group combinations, others met nutrient targets through fortification. Greater diversity of healthy food groups, often driven by multiple daily meals, was associated with higher Modified GDQS-Menu scores. However, achieving this diversity can be difficult in low-resource settings due to cost, availability, and infrastructure challenges. Within these resource-constrained contexts, fortification and biofortification can serve as critical strategies

for enhancing nutritional quality. Nevertheless, reliance on fortified food vehicles, such as biscuits or other processed snack items, entails trade-offs, as these foods may also contribute higher sugar content. This pattern underscores the importance of shifting toward greater dietary diversity as food system capacities grow.

Additionally, while GDP per capita had a modest relationship with Modified GDQS-Menu scores, substantial variability remained among countries with similar incomes. These findings suggest the value of identifying *aspirational peers* – programs operating in comparable socioeconomic and food system contexts that perform well on nutrient adequacy and GDQS metrics. Learning from peer programs can help countries identify feasible, context-specific pathways for improving school meal quality aligned with cultural preferences, food environments, and national nutrition goals.⁽⁴⁵⁾ Designing and implementing school meal guidelines or policies is critical for developing these context-specific strategies. Countries with school meal policies tend to have broader program coverage and higher-quality meals.^{(38),(46)} Evidence-based guidance, supported through technical cooperation with *aspirational peers*, can help countries design and tailor school meal standards that address the unique complexities of country-specific DBM landscapes.⁽³⁸⁾

Next Steps

Future research could examine differences between planned and as-delivered menu quality to better reflect real-world program implementation. Research could also use longitudinal data to track trends in school meal quality over time, particularly as food systems evolve. Additionally, research should position school meals within children's broader consumption patterns. Without data on what children consume outside of school, it is difficult to determine how well their daily nutritional needs are being met and how school meals can best complement broader dietary behaviors. Future studies can also adopt a broader lens, extending beyond a focus on nutrition to simultaneously consider the environmental sustainability of school meals, given the role that school meal programs can play in transforming food systems. Context-specific procurement models, such as Home-Grown School Feeding and Farm-to-School, and the incorporation of underutilized, nutrient-rich Indigenous foods could be explored as pathways for meaningful, multi-sectoral food systems change. Within this realm, research should explore trade-offs among nutritional quality, cost, and environmental impact, and identify pathways for meal optimization. Notably, this

analysis can include an application of the forthcoming GDQS Environmentally Sustainable Healthy-Meal and -Menu metrics to evaluate food baskets on both nutritional and sustainability dimensions.⁽⁴⁷⁾

Limitations

Despite standardization of the GSSMP, different menu formats, terminologies, cut-off points, and irregular data complicated comparability between countries. Meal types and age ranges also vary across countries, complicating nutritional adequacy assessments for children at different developmental stages. Only a subset of programs provided sufficiently detailed quantitative menus for inclusion in this review, limiting the representativeness of the sample. Programs with available quantitative menus may also be more established or better resourced, and findings, therefore, may not be generalizable to all school meal programs. Additionally, subnational program variations were not captured. The assumptions of comparability by week, complete biofortification, common variants, and cooked status may not hold true in all cases. Most importantly, this data provides a cross-sectional view of a dynamic school meal landscape.

Conclusion

This study addresses a critical evidence gap in the nutritional quality of school meal programs by combining nutrient composition analysis with GDQS-Meal and -Menu metrics, offering a holistic assessment of both nutrient adequacy and overall meal quality. School meals contribute significantly to children's daily nutrient needs, and most menus achieve high nutritional quality. However, significant variability across countries, age groups, and meal types highlights opportunities for improvement through age- and context-specific strategies. While achieving greater diversity, especially of healthy foods, can be challenging in low-resource settings, fortification remains a promising pathway for enhancing meal quality. Substantial variability among countries with similar income levels also suggests learning from well-performing peer programs to tailor feasible, context-specific improvements. Future research should examine differences between planned and as-delivered menu quality, trends over time, and children's overall diets. Research should also explore trade-offs among nutritional quality, cost, and environmental sustainability to inform actionable, multi-sectoral improvements in school meal programs.

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Table 1: Characteristics of School Meal Programs by Country

Country	Region	Income Group ¹	GDP per Capita (\$US)	Program(s)	Menu Source ²	Child Age ³	Meal Type ⁴	Menu Language	Meal Days
Afghanistan	South Asia	LIC	416	Provision of Fortified Biscuits (FBs), Take-Home Rations (THRs – Fortified Veg. Oil), Bread+ (Nutritious Bread), Incentives for Secondary Girls, and Deworming Campaign	Literature Review	Primary	Snack	English	6
Armenia	Europe & Central Asia	UMIC	8,053	"School Feeding" National Program of Republic of Armenia	GSSMP	Preschool; Primary	Lunch	Armenian & English	5
Bangladesh	South Asia	LMIC	2,551	School Feeding Programme in Poverty-Prone Areas	Literature Review	Preschool; Primary	Snack	English	6
Belize	Latin America & Caribbean	UMIC	7,460	National Healthy Start School Feeding Program	GSSMP	Primary	Breakfast; Lunch	English	4
Cambodia	East Asia & Pacific	LMIC	2,430	Home-Grown School Feeding Program	GSSMP	Preschool; Primary	Lunch	English	6
Canada	North America	HIC	53,431	Province/Territory-level school meal programs	GSSMP	Primary; Secondary	Breakfast; Lunch	English	5
Costa Rica	Latin America & Caribbean	UMIC	16,942	Food and Nutrition Program for Schoolchildren and Adolescents (Programa de Alimentación y Nutrición del Escolar y el Adolescente)	GSSMP	Preschool; Primary; Secondary	Breakfast; Lunch	Spanish	5
Cyprus	Europe & Central Asia	HIC	36,551	Lunch for Pupils Attending All-Day Optional Primary Schools	GSSMP	Preschool; Primary	Lunch	Greek	5
Djibouti	Middle East	LMIC	3,555	School Catering (Restauration scolaire)	Literature	Preschool;	Lunch;	English	5

Country	Region	Income Group ¹	GDP per Capita (\$US)	Program(s)	Menu Source ²	Child Age ³	Meal Type ⁴	Menu Language	Meal Days
	& North Africa				Review	Primary; Secondary	Snack		
eSwatini	Sub-Saharan Africa	LMIC	3,611	Home-Grown School Feeding (pilot program)	Literature Review	Preschool; Primary; Secondary	Lunch	English	5
Guatemala	Latin America & Caribbean	UMIC	5,763	School Feeding Program (Programa de Alimentación Escolar)	Literature Review	Preschool; Primary	Lunch	English	7
Honduras	Latin America & Caribbean	LMIC	3,232	National School Feeding Program (Programa Nacional de Alimentación Escolar)	GSSMP	Preschool; Primary	Breakfast	Spanish	5
India	South Asia	LMIC	2,481	Pradhan Mantri Poshan Shakti Nirman (PM POSHAN) Scheme	Literature Review	Primary; Secondary	Lunch	English	5
Latvia	Europe & Central Asia	HIC	22,503	Free meals (Brīvpusdienas)	GSSMP	Primary	Lunch	Latvian	5
Lesotho	Sub-Saharan Africa	LMIC	916	Bokamoso Ba Bana (3B) School Feeding Project; WFP Pre-Primary School Feeding Project (ECCD)	GSSMP	Preschool; Primary	Lunch	English	5
Moldova	Europe & Central Asia	UMIC	6,729	School Food Program (школьного питания)	GSSMP	Primary	Breakfast; Lunch; Snack	Romanian	5
Mongolia	East Asia &	UMIC	5,839	School Lunch Program	GSSMP	Primary	Lunch	Mongolian	5

Country	Region	Income Group ¹	GDP per Capita (\$US)	Program(s)	Menu Source ²	Child Age ³	Meal Type ⁴	Menu Language	Meal Days
	Pacific								
Nepal	South Asia	LMIC	1,378	Food for Education Project (FFEP)	GSSMP	Preschool; Primary	Lunch	English	6
New Zealand	East Asia & Pacific	HIC	48,281	Ka Ora, Ka Ako (Healthy School Lunches)	GSSMP	Primary; Secondary	Lunch	English	5
Poland	Europe & Central Asia	HIC	22,057	Meal at School and at Home	GSSMP	Primary; Secondary	Lunch	Polish	5
Republic of Congo	Sub-Saharan Africa	LMIC	2,478	McGovern-Dole School Canteen (Cantine scolaire McGovern-Dole)	GSSMP	Preschool; Primary	Lunch	French	5
South Africa	Sub-Saharan Africa	UMIC	6,023	National School Nutrition Programme (NSNP)	GSSMP	Primary; Secondary	Breakfast; Lunch	English	5
South Sudan	Sub-Saharan Africa	LIC	1,080	School Feeding Program	Literature Review	Primary; Secondary	Lunch	English	5
Sri Lanka	South Asia	LMIC	3,828	School Nutrition Programme	GSSMP	Primary	Breakfast	English	5
Tunisia	Middle East & North Africa	LMIC	3,978	National School Feeding Program (Programme National d'Alimentation Scolaire)	GSSMP	Primary; Secondary	Lunch	French	4
USA	North America	HIC	82,769	National School Lunch Program; School Breakfast Program	Literature Review	Primary; Secondary	Lunch	English	5
Uruguay	Latin America &	HIC	22,798	School Feeding Program (Programa de Alimentación Escolar)	GSSMP	Preschool; Primary;	Lunch; Snack	Spanish	5

Country	Region	Income Group ¹	GDP per Capita (\$US)	Program(s)	Menu Source ²	Child Age ³	Meal Type ⁴	Menu Language	Meal Days
	Caribbean					Secondary			
Uzbekistan	Europe & Central Asia	LMIC	2,850	Providing meals to primary school students in the Aral Sea region (Обеспечение питанием учащихся начальных школ региона Приаралья)	GSSMP	Primary	Breakfast	English/Uzbek	5
Yemen	Middle East & North Africa	LIC	477	School Feeding Program (Biscuits)	Literature Review	Primary	Snack	English	5

¹ LIC = Low Income Country, LMIC = Lower-Middle Income Country, UMIC = Upper-Middle Income Country, HIC = High Income Country

² GSSMP = Global Survey of School Meal Programs

³ Programs may serve meals to additional child age groups; however, only those with identified menus for specific age groups are included.

⁴ Programs may serve additional meal types; however, only meal types with identified menus are included.

Table 2: Mean Daily Nutritional Composition of School Meals by Country Averaged Across Age Groups with Percentage of Nutrient Reference Value (NRV) Achieved

Country	Energy (kcal), [NRV %]	Nutrients																
		Protein (g), [NRV%]	Carbohy- drate (g), [NRV%]	Fiber (g), [NRV %]	Total Sugar (g), [NRV%]	Total Fat (g), [NRV %]	Saturate- d Fat (g), [NRV%]	Calcium (mg), [NRV%]	Iron (mg), [NRV%]	Sodium (mg), [NRV %]	Vitamin A (µg) -RAE, [NRV%]	Thiamin (mg), [NRV%]	Riboflavi- n (mg), [NRV%]	Niacin (mg), [NRV%]	Vitamin B6 (mg), [NRV%]	Folate DFE (µg), [NRV%]	Vitamin B12 (µg), [NRV%]	Vitamin C (mg), [NRV%]
Afghanistan	440.00 (28.0%)	9.00 (34.0%)	89.60 (68.9%)	2.25 (8.4%)	14.67 (37.3%)	15.00 (28.6%)	0.00 (0.0%)	0.00 (0.0%)	11.00 (169.2%)	0.00 (0.0%)	250.00 (88.5%)	0.49 (98.0%)	0.68 (97.1%)	5.87 (97.8%)	1.10 (146.7%)	0.00 (0.0%)	2.20 (220.0%)	20.00 (61.5%)
Armenia	449.08 (33.0%)	17.92 (89.8%)	65.99 (50.8%)	11.49 (47.6%)	15.39 (45.3%)	13.37 (29.5%)	3.17 (21.0%)	599.09 (100.0%)	4.92 (87.0%)	1264.75 (63.2%)	48.31 (19.3%)	0.48 (102.2%)	0.36 (58.1%)	6.17 (107.4%)	0.30 (47.0%)	174.61 (152.0%)	0.21 (23.1%)	17.48 (70.6%)
Bangladesh	337.50 (24.8%)	9.38 (47.0%)	68.73 (52.9%)	1.72 (7.1%)	11.25 (33.1%)	11.25 (24.8%)	0.00 (0.0%)	0.00 (0.0%)	8.25 (146.0%)	0.00 (0.0%)	187.50 (74.9%)	0.38 (80.2%)	0.52 (84.4%)	4.50 (78.4%)	0.75 (118.2%)	1020.00 (887.8%)	0.38 (41.4%)	15.00 (60.6%)
Belize	1124.70 (71.6%)	46.19 (174.3%)	152.58 (117.4%)	16.34 (61.1%)	48.34 (123.0%)	39.24 (74.9%)	11.31 (64.8%)	397.26 (58.4%)	8.81 (135.5%)	681.47 (34.1%)	1222.47 (432.7%)	1.00 (200.5%)	0.85 (121.4%)	15.96 (266.0%)	1.24 (164.7%)	361.89 (268.1%)	0.47 (47.0%)	186.30 (573.2%)
Cambodia	164.39 (12.1%)	4.96 (24.8%)	14.46 (11.1%)	1.11 (4.6%)	1.93 (5.7%)	10.17 (22.4%)	5.78 (38.2%)	80.08 (13.4%)	2.07 (36.6%)	488.75 (24.4%)	133.85 (53.4%)	0.12 (25.7%)	0.13 (20.8%)	1.21 (21.0%)	0.26 (41.0%)	54.63 (47.5%)	0.61 (65.8%)	12.46 (50.3%)
Canada	1305.62	59.78	184.76	24.51	57.44	40.95	16.23 (72.3%)	913.21	11.57	1112.86	405.22 (105.5%)	1.47	1.55 (164.8%)	11.10	1.35	291.91	2.60	144.33

	(64.9%)	(176.8%)	(142.1%)	(85.5%)	(114.8%)	(60.7%)		(111.0%)	(128.4%)	(55.6%)		(226.6%)		(139.0%)	(134.7%)	(162.7%)	(193.9%)	(299.6%)
Costa Rica	448.52 (25.6%)	19.84 (74.8%)	65.45 (50.3%)	7.81 (29.1%)	16.60 (40.4%)	12.80 (21.5%)	5.90 (30.4%)	155.43 (22.9%)	3.00 (39.3%)	582.06 (29.1%)	163.79 (51.1%)	0.53 (91.6%)	0.36 (47.7%)	4.64 (66.0%)	0.64 (77.2%)	150.08 (96.4%)	0.58 (50.3%)	41.71 (112.7%)
Cyprus	582.79 (43.0%)	33.01 (166.2%)	66.33 (51.0%)	12.61 (52.5%)	24.85 (73.4%)	22.75 (50.2%)	5.85 (38.7%)	411.64 (68.8%)	5.57 (99.0%)	635.00 (31.8%)	1055.13 (421.4%)	0.67 (141.6%)	0.85 (137.8%)	8.34 (145.6%)	0.99 (157.3%)	216.05 (189.0%)	1.06 (115.8%)	71.90 (292.0%)
Djibouti	772.10 (46.7%)	20.67 (81.6%)	89.31 (68.7%)	6.03 (22.9%)	13.61 (32.5%)	21.09 (39.1%)	12.42 (69.6%)	330.72 (47.4%)	8.24 (116.9%)	1572.14 (78.6%)	580.83 (188.9%)	0.39 (68.7%)	0.96 (126.7%)	6.90 (101.0%)	0.56 (71.1%)	149.73 (105.1%)	1.13 (102.1%)	34.00 (104.1%)
eSwatini	633.08 (39.4%)	19.21 (79.6%)	104.53 (80.4%)	9.21 (35.6%)	5.24 (13.1%)	14.84 (27.7%)	2.69 (15.1%)	94.30 (13.8%)	2.76 (40.2%)	1081.62 (54.1%)	89.36 (29.7%)	0.33 (59.4%)	0.37 (49.9%)	1.65 (24.5%)	0.43 (55.9%)	141.52 (102.6%)	0.56 (51.3%)	38.62 (122.4%)
Guatemala	474.94 (34.9%)	12.39 (62.1%)	75.84 (58.3%)	4.71 (19.5%)	20.53 (60.4%)	14.39 (31.8%)	9.67 (64.0%)	168.85 (28.2%)	2.51 (44.5%)	104.46 (5.2%)	42.52 (17.0%)	0.26 (55.2%)	0.26 (42.9%)	2.09 (36.3%)	0.18 (28.1%)	108.11 (94.1%)	0.42 (45.9%)	3.00 (12.1%)
Honduras	273.71 (20.1%)	8.64 (43.3%)	54.34 (41.8%)	9.85 (40.8%)	5.43 (16.0%)	3.82 (8.4%)	1.93 (12.8%)	287.48 (48.0%)	3.61 (63.9%)	46.22 (2.3%)	46.59 (18.6%)	0.42 (88.7%)	0.41 (66.6%)	3.14 (54.7%)	0.08 (12.6%)	145.48 (126.6%)	0.43 (46.8%)	0.61 (2.5%)
India	623.86 (30.9%)	15.33 (45.6%)	116.89 (89.9%)	5.80 (20.2%)	5.20 (10.5%)	10.04 (15.3%)	5.14 (23.6%)	89.36 (10.9%)	2.82 (31.7%)	56.49 (2.8%)	73.43 (19.3%)	0.23 (35.8%)	0.18 (19.7%)	3.38 (42.6%)	0.41 (40.9%)	85.12 (47.6%)	0.10 (7.8%)	13.77 (28.5%)
Latvia	645.45 (41.1%)	31.63 (119.4%)	81.72 (62.9%)	14.28 (53.4%)	28.00 (71.3%)	24.22 (46.2%)	7.41 (42.4%)	369.48 (54.3%)	6.20 (95.4%)	541.37 (27.1%)	917.62 (324.8%)	0.41 (82.0%)	0.75 (107.7%)	8.52 (142.0%)	1.09 (145.6%)	145.78 (108.0%)	1.34 (133.6%)	32.34 (99.5%)

Lesotho	945.63 (69.0%)	28.28 (147.7%)	123.97 (95.4%)	9.09 (37.6%)	2.67 (7.6%)	19.69 (43.9%)	9.55 (63.1%)	364.94 (62.9%)	14.17 (248.7%)	271.97 (13.6%)	646.55 (260.4%)	1.08 (225.8%)	1.54 (248.9%)	14.40 (250.9%)	0.81 (128.8%)	594.73 (501.7%)	3.04 (346.8%)	36.99 (148.9%)
Moldova	882.76 (64.9%)	43.94 (220.2%)	120.52 (92.7%)	15.38 (63.7%)	52.06 (153.2%)	27.44 (60.6%)	8.76 (58.0%)	661.89 (110.5%)	5.82 (102.9%)	1255.68 (62.8%)	500.85 (199.9%)	0.93 (195.5%)	1.01 (163.6%)	11.32 (197.3%)	1.11 (175.5%)	217.12 (189.0%)	1.80 (195.7%)	77.27 (312.1%)
Mongolia	567.09 (36.1%)	35.50 (134.0%)	49.07 (37.7%)	4.98 (18.6%)	12.37 (31.5%)	26.02 (49.7%)	12.82 (73.4%)	222.04 (32.7%)	4.87 (74.9%)	1198.19 (59.9%)	1375.25 (486.8%)	0.34 (68.0%)	0.84 (120.6%)	7.79 (129.8%)	0.74 (98.1%)	116.86 (86.6%)	10.39 (1039.0%)	42.27 (130.1%)
Nepal	225.97 (16.6%)	4.73 (23.7%)	29.20 (22.5%)	2.71 (11.2%)	1.29 (3.8%)	10.41 (23.0%)	8.25 (54.6%)	44.98 (7.5%)	2.17 (38.4%)	785.36 (39.3%)	120.47 (48.1%)	0.19 (40.5%)	0.08 (13.5%)	1.67 (29.0%)	0.20 (31.8%)	129.97 (113.1%)	0.00 (0.0%)	7.86 (31.8%)
New Zealand	491.51 (25.0%)	23.02 (69.8%)	51.23 (39.4%)	5.77 (20.3%)	10.40 (21.6%)	22.13 (33.5%)	8.25 (37.5%)	106.31 (13.2%)	3.09 (36.0%)	460.32 (23.0%)	193.87 (56.0%)	0.47 (74.1%)	0.33 (36.3%)	7.07 (91.8%)	0.58 (60.0%)	97.23 (56.9%)	0.65 (49.6%)	28.35 (64.4%)
Poland	671.48 (33.7%)	32.04 (98.8%)	90.63 (69.7%)	19.40 (67.8%)	27.27 (54.6%)	23.96 (35.6%)	4.30 (19.0%)	454.32 (55.6%)	9.86 (113.5%)	644.53 (32.2%)	886.86 (216.4%)	0.50 (76.7%)	1.09 (116.1%)	11.74 (154.9%)	1.17 (120.8%)	231.08 (128.3%)	0.64 (50.6%)	113.97 (251.8%)
Republic of the Congo	420.30 (30.9%)	13.57 (68.0%)	48.51 (37.3%)	2.80 (11.6%)	2.45 (7.2%)	18.94 (41.8%)	12.81 (84.8%)	141.60 (23.6%)	1.81 (32.0%)	2604.20 (130.2%)	25.60 (10.2%)	0.16 (33.8%)	0.15 (24.4%)	2.98 (51.9%)	0.28 (44.1%)	32.70 (28.5%)	2.68 (291.6%)	5.68 (22.9%)
South Africa	274.74 (14.4%)	12.90 (39.5%)	45.62 (35.1%)	5.14 (18.2%)	8.88 (18.1%)	4.93 (7.6%)	0.95 (4.4%)	130.39 (16.0%)	2.34 (29.0%)	490.44 (24.5%)	216.62 (57.3%)	0.24 (40.8%)	0.24 (27.1%)	3.49 (46.8%)	0.31 (32.2%)	116.17 (71.2%)	1.03 (76.9%)	20.53 (43.5%)

South Sudan	538.78 (27.8%)	13.66 (42.2%)	96.88 (74.5%)	8.92 (31.4%)	2.69 (5.6%)	13.48 (20.9%)	8.72 (40.5%)	20.16 (2.5%)	3.85 (45.6%)	1555.72 (77.8%)	5.28 (1.5%)	0.36 (57.5%)	0.11 (12.3%)	3.31 (43.7%)	0.44 (46.5%)	67.80 (39.9%)	0.00 (0.0%)	0.36 (0.8%)
Sri Lanka	565.94 (36.0%)	21.87 (82.5%)	108.32 (83.3%)	7.56 (28.3%)	5.49 (14.0%)	4.76 (9.1%)	1.08 (6.2%)	139.32 (20.5%)	8.05 (123.8%)	52.66 (2.6%)	109.32 (38.7%)	0.96 (192.0%)	0.33 (46.9%)	6.98 (116.3%)	0.53 (70.9%)	558.28 (413.5%)	0.95 (94.6%)	48.75 (150.0%)
Tunisia	354.19 (18.3%)	21.80 (67.3%)	32.74 (25.2%)	6.90 (24.2%)	5.64 (11.7%)	16.19 (25.1%)	3.39 (15.8%)	141.90 (17.7%)	4.01 (47.1%)	696.58 (34.8%)	472.76 (130.4%)	0.19 (29.9%)	0.38 (42.2%)	6.48 (85.6%)	0.60 (62.8%)	92.66 (54.5%)	0.36 (28.3%)	27.94 (64.7%)
United States	1063.28 (54.7%)	51.43 (158.0%)	160.14 (123.2%)	17.73 (62.4%)	75.27 (155.6%)	28.60 (43.9%)	11.83 (54.8%)	1046.46 (131.4%)	8.43 (98.9%)	1295.05 (64.8%)	606.45 (167.1%)	1.26 (201.5%)	1.83 (203.8%)	11.19 (146.4%)	1.37 (143.7%)	221.60 (130.1%)	3.54 (277.4%)	105.66 (238.0%)
Uruguay	901.70 (56.2%)	38.19 (158.4%)	122.69 (94.4%)	12.83 (49.6%)	52.81 (131.6%)	31.04 (58.0%)	10.69 (59.9%)	574.14 (83.8%)	5.78 (84.3%)	1339.92 (67.0%)	552.03 (183.1%)	0.73 (131.9%)	0.90 (120.8%)	10.70 (159.1%)	0.97 (127.2%)	217.06 (157.4%)	2.85 (260.5%)	70.91 (224.7%)
Uzbekistan	575.12 (36.6%)	10.09 (38.1%)	82.95 (63.8%)	6.38 (23.8%)	36.43 (92.7%)	24.53 (46.8%)	13.77 (78.9%)	65.57 (9.6%)	1.71 (26.4%)	857.10 (42.9%)	190.62 (67.5%)	0.30 (60.4%)	0.29 (40.9%)	2.30 (38.3%)	0.14 (19.2%)	58.81 (43.6%)	0.18 (17.8%)	4.94 (15.2%)
Yemen	440.00 (28.0%)	9.00 (34.0%)	89.60 (68.9%)	2.25 (8.4%)	14.67 (37.3%)	15.00 (28.6%)	0.00 (0.0%)	0.00 (0.0%)	11.00 (169.2%)	0.00 (0.0%)	250.00 (88.5%)	0.49 (98.0%)	0.68 (97.1%)	5.87 (97.8%)	1.10 (146.7%)	0.00 (0.0%)	2.20 (220.0%)	20.00 (61.5%)

Table 3: Modified GDQS-Menu Scores by Country, Averaged Across Age Groups

Country	Sub-Metrics				Modified GDQS- Menu Score
	Menu+	Menu-	Fortification & Biofortification	Food Group Diversity	
Afghanistan	0.00	14.46	8.43	0.00	22.89
Armenia	7.86	13.93	0.00	12.05	33.84
Bangladesh	0.00	14.46	19.28	0.00	33.73
Belize	17.74	12.95	2.41	18.07	51.17
Cambodia	7.40	16.00	7.23	10.84	41.47
Canada	11.98	16.80	6.02	14.46	49.27
Costa Rica	10.13	13.68	0.00	15.66	39.46
Cyprus	12.17	15.32	0.00	14.46	41.94
Djibouti	6.75	12.84	3.61	2.41	25.62
eSwatini	10.40	14.46	2.41	6.02	33.29
Guatemala	9.35	13.27	10.84	6.02	39.49
Honduras	2.41	13.84	7.23	1.20	24.69
India	9.39	13.80	6.02	12.05	41.26
Latvia	14.13	14.00	1.20	19.28	48.61
Lesotho	8.26	14.46	6.62	4.22	33.55
Moldova	18.54	12.52	3.61	19.28	53.95
Mongolia	8.03	14.14	0.00	15.66	37.83
Nepal	11.55	14.46	10.84	7.23	44.08
New Zealand	7.34	14.14	2.41	15.66	39.56
Poland	16.16	13.11	0.00	18.08	47.34
Republic of Congo	9.30	14.46	1.20	3.61	28.58
South Africa	5.61	14.46	8.43	12.05	40.55
South Sudan	8.23	15.90	3.61	3.61	31.37
Sri Lanka	13.07	14.46	3.61	9.64	40.78
Tunisia	10.40	14.73	0.00	10.84	35.97
United States	17.11	13.69	12.05	16.87	59.72
Uruguay	14.96	12.16	2.41	20.48	50.01
Uzbekistan	7.53	12.11	0.00	7.23	26.86
Yemen	0.00	14.46	8.43	0.00	22.89

Table 4. Mean Daily Grams Consumed per GDQS Food Group (Global and Country-Level)

GDQS-Meal/Menu Categories		Global	Country																												
Food Group Category	Food Group		Afghanistan	Armenia	Bangladesh	Belize	Cambodia	Canada	Costa Rica	Cyprus	Djibouti	eSwatini	Guatemala	Honduras	India	Latvia	Lesotho	Moldova	Mongolia	Nepal	New Zealand	Poland	Republic of Congo	South Africa	South Sudan	Sri Lanka	Tunisia	United States	Uruguay	Uzbekistan	Yemen
Healthy	Other Fruits	59.2	0.0	47.7	0.0	240.0	2.0	257.5	90.6	0.0	0.0	0.0	0.0	0.0	13.8	81.2	0.0	133.6	24.0	0.0	0.0	107.3	0.0	182.0	0.0	50.0	0.0	131.4	256.0	100.0	0.0
	Cruciferous Vegetables	39.0	0.0	50.0	0.0	45.4	0.0	180.0	40.8	83.3	0.0	100.0	0.0	0.0	7.1	57.0	60.0	80.0	55.2	20.0	38.9	116.0	0.0	70.0	0.0	0.0	0.0	78.5	48.3	0.0	0.0
	Other Vegetables	48.7	0.0	16.5	0.0	154.3	33.0	169.6	51.1	275.4	0.0	8.0	19.8	0.0	28.5	92.6	0.0	121.7	49.4	0.0	53.0	93.0	0.0	25.6	0.0	30.0	59.6	74.1	57.1	0.0	0.0
	Legumes	40.5	0.0	40.0	0.0	49.6	0.0	108.2	95.9	120.0	26.7	40.0	46.4	0.0	28.4	22.5	30.0	19.9	35.0	20.0	153.8	41.7	40.0	37.5	30.0	38.4	15.0	80.9	54.7	0.0	0.0
	Deep Orange Tubers	23.6	0.0	10.0	0.0	79.3	10.0	0.0	21.6	117.2	0.0	0.0	0.0	0.0	10.0	50.2	0.0	20.8	19.0	20.0	30.0	81.5	0.0	70.0	0.0	0.0	70.0	36.0	39.2	0.0	0.0
	Whole Grains	37.5	0.0	50.0	0.0	0.0	0.0	192.2	0.0	40.0	0.0	0.0	278.1	0.0	0.0	55.7	0.0	56.8	0.0	0.0	77.7	57.5	0.0	0.0	120.0	0.0	0.0	62.2	61.5	36.0	0.0
	Liquid Oils	6.9	0.0	7.6	0.0	20.0	5.0	0.0	3.4	12.0	15.0	7.5	7.3	0.0	4.0	9.0	10.5	6.2	15.4	10.0	4.1	15.8	15.0	2.0	10.0	0.0	7.5	1.2	10.7	0.4	0.0
	Poultry and Game Meats	30.4	0.0	50.0	0.0	78.0	0.0	42.5	54.2	80.0	0.0	0.0	0.0	0.0	0.0	73.5	0.0	94.5	60.0	0.0	59.7	63.3	0.0	53.4	0.0	0.0	70.0	53.3	50.0	0.0	0.0
	Eggs	15.0	0.0	25.0	0.0	27.5	20.0	68.8	43.0	0.0	0.0	50.0	0.0	0.0	44.0	10.2	42.0	0.0	0.0	0.0	0.0	10.0	0.0	0.0	0.0	40.0	35.0	0.0	10.1	9.1	0.0
	Citrus Fruits	20.2	0.0	0.0	0.0	120.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.0	0.0	70.8	0.0	0.0	90.3	50.0	0.0	0.0	0.0	0.0	0.0	107.4	140.0	0.0	0.0
	Deep Orange Fruits	13.4	0.0	0.0	0.0	160.0	0.0	0.0	83.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	50.0	0.0	0.0	95.0	0.0	0.0
	Dark Green Leafy	9.7	0.0	0.0	0.0	18.1	44.0	0.0	3.2	41.7	0.0	0.0	0.0	0.0	4.5	1.1	0.0	5.1	1.0	20.0	17.1	20.8	0.0	0.0	0.0	30.0	28.8	45.6	1.1	0.0	0.0

	Vegetables																															
	Deep Orange Vegetables	21.4	0.0	0.0	0.0	127.0	30.0	42.5	12.9	0.0	0.0	0.0	0.0	0.0	0.0	45.3	0.0	43.1	16.0	20.0	36.0	125.0	0.0	70.0	0.0	30.0	10.0	0.0	12.5	0.0	0.0	
	Low Fat Dairy	84.3	0.0	0.0	0.0	240.0	0.0	412.5	80.3	200.0	0.0	0.0	13.0	96.0	25.0	165.8	0.0	312.5	66.7	0.0	64.1	69.0	0.0	225.0	0.0	0.0	0.0	373.7	83.0	18.0	0.0	
	Fish and Shellfish	18.6	0.0	0.0	0.0	0.0	20.0	42.5	0.0	60.0	0.0	0.0	0.0	0.0	0.0	50.0	53.0	68.4	0.0	0.0	0.0	80.0	30.0	50.0	0.0	26.7	0.0	0.0	60.0	0.0	0.0	
	Nuts and Seeds	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	7.1	0.0	0.0	6.2	0.3	0.0	0.4	10.0	0.0	0.0	0.0	0.0	0.0	0.0	32.0	0.0	5.0	0.0
Unhealthy in Excess	High Fat Dairy	14.4	0.0	20.0	0.0	32.9	0.0	66.3	30.0	22.5	0.0	0.0	0.0	0.0	0.0	28.0	0.0	6.9	20.0	0.0	0.0	11.2	0.0	0.0	0.0	0.0	0.0	162.7	16.7	0.0	0.0	
	Red Meat	25.1	0.0	0.0	0.0	85.0	20.0	39.7	82.5	60.0	0.0	0.0	0.0	0.0	0.0	60.0	0.0	76.8	56.4	0.0	83.2	0.0	0.0	0.0	0.0	0.0	70.0	45.4	50.0	0.0	0.0	
Unhealthy	Sweets and Ice Cream	18.6	100.0	12.0	75.0	1.5	1.0	7.5	4.0	0.0	13.3	0.0	17.2	0.1	10.0	5.0	0.0	13.2	4.4	0.0	2.4	7.5	0.0	0.5	0.0	0.0	0.0	113.0	23.4	27.8	100.0	
	Refined Grains and Baked Goods	97.9	0.0	95.0	0.0	160.7	115.0	0.0	90.3	50.5	190.0	150.0	78.3	48.0	387.5	29.6	205.0	46.7	44.0	80.0	129.5	48.3	150.0	60.0	120.0	310.0	50.0	32.9	133.7	35.0	0.0	
	White Roots and Tubers	27.0	0.0	50.0	0.0	54.4	0.0	0.0	85.0	45.0	0.0	0.0	0.0	0.0	13.5	95.3	0.0	82.8	32.0	0.0	57.3	97.1	0.0	0.0	0.0	0.0	60.0	0.0	111.7	0.0	0.0	
	Juice	14.1	0.0	0.0	0.0	128.3	0.0	0.0	4.7	150.0	0.0	0.0	0.0	0.0	0.0	3.0	0.0	1.2	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	119.5	0.0	1.0	0.0	
	Processed Meat	2.4	0.0	0.0	0.0	0.0	0.0	36.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	17.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.0	0.0	0.0	
Not Classified	Not Classified	67.6	0.0	2.0	0.0	50.5	69.4	21.9	58.2	120.7	3.0	2.0	0.0	120.2	5.3	253.5	0.0	10.5	468.1	2.0	46.7	146.6	5.0	1.4	4.0	0.0	11.7	81.7	245.3	230.0	0.0	

Figure 1: Distribution of Nutrient Reference Value (NRV) Percentage Achieved by Each Nutrient Overall and Disaggregated by Age and Meal Type.



Figure 2: GDQS-Meal and -Menu Scores by Age and Meal Type.

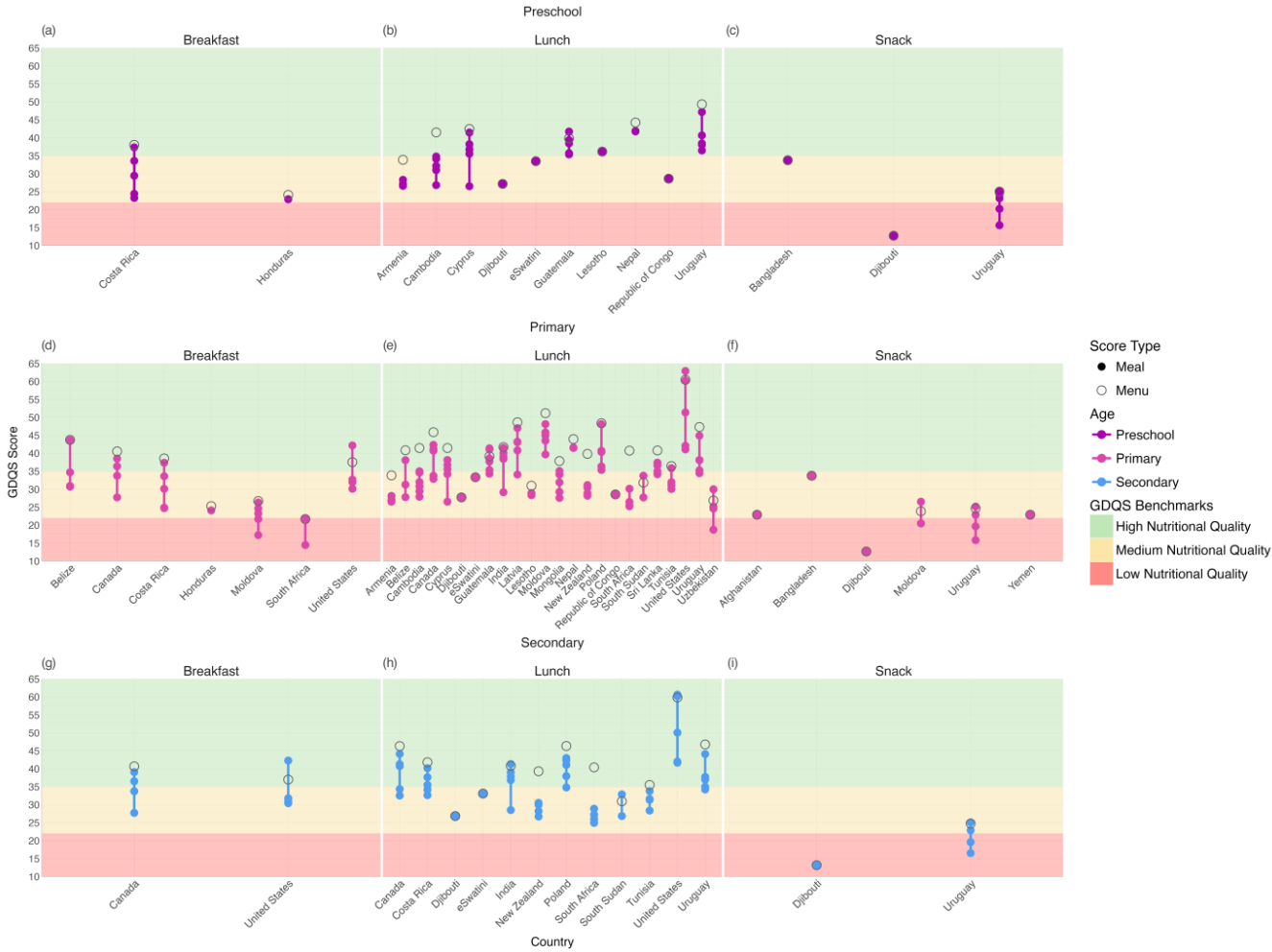


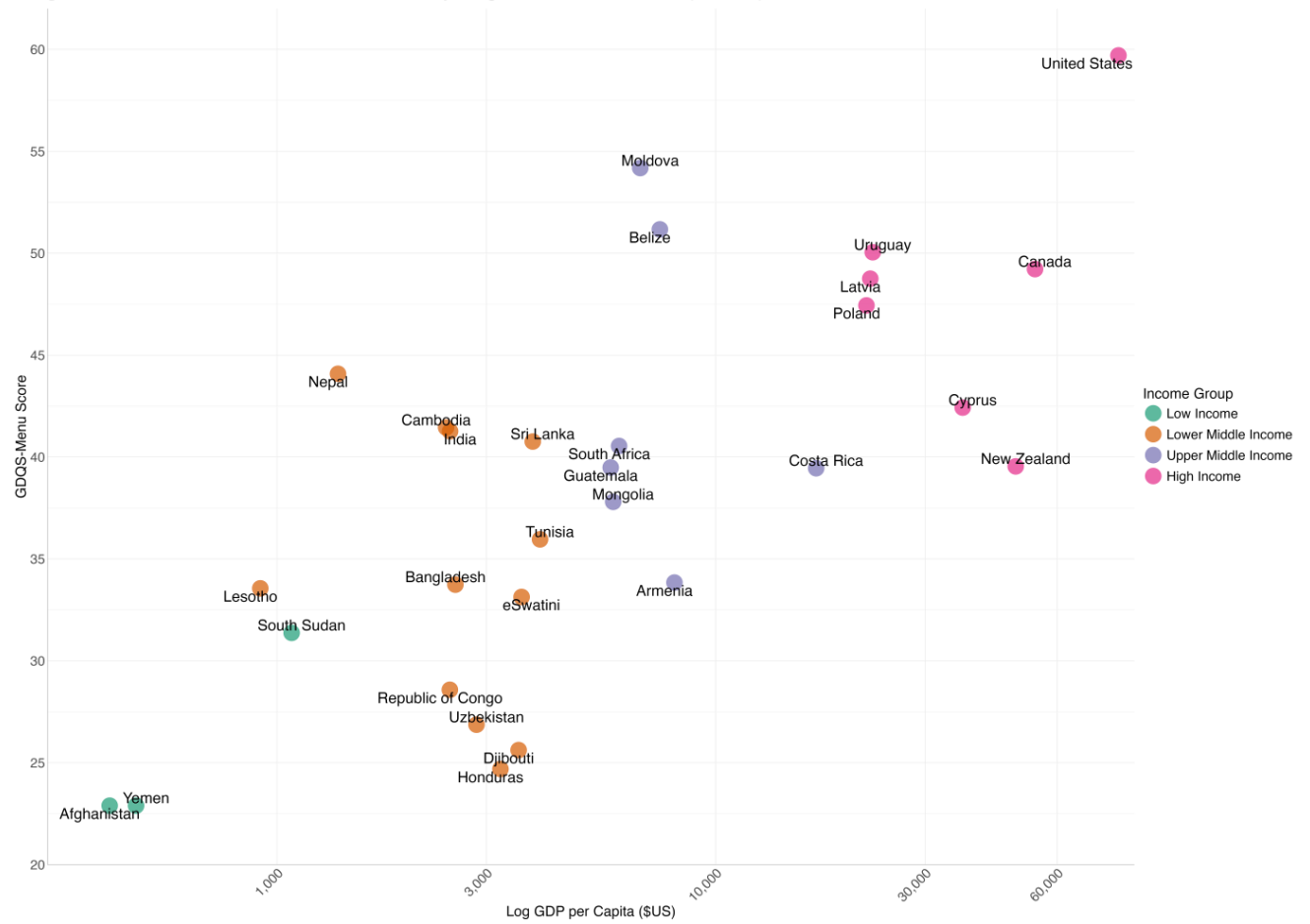
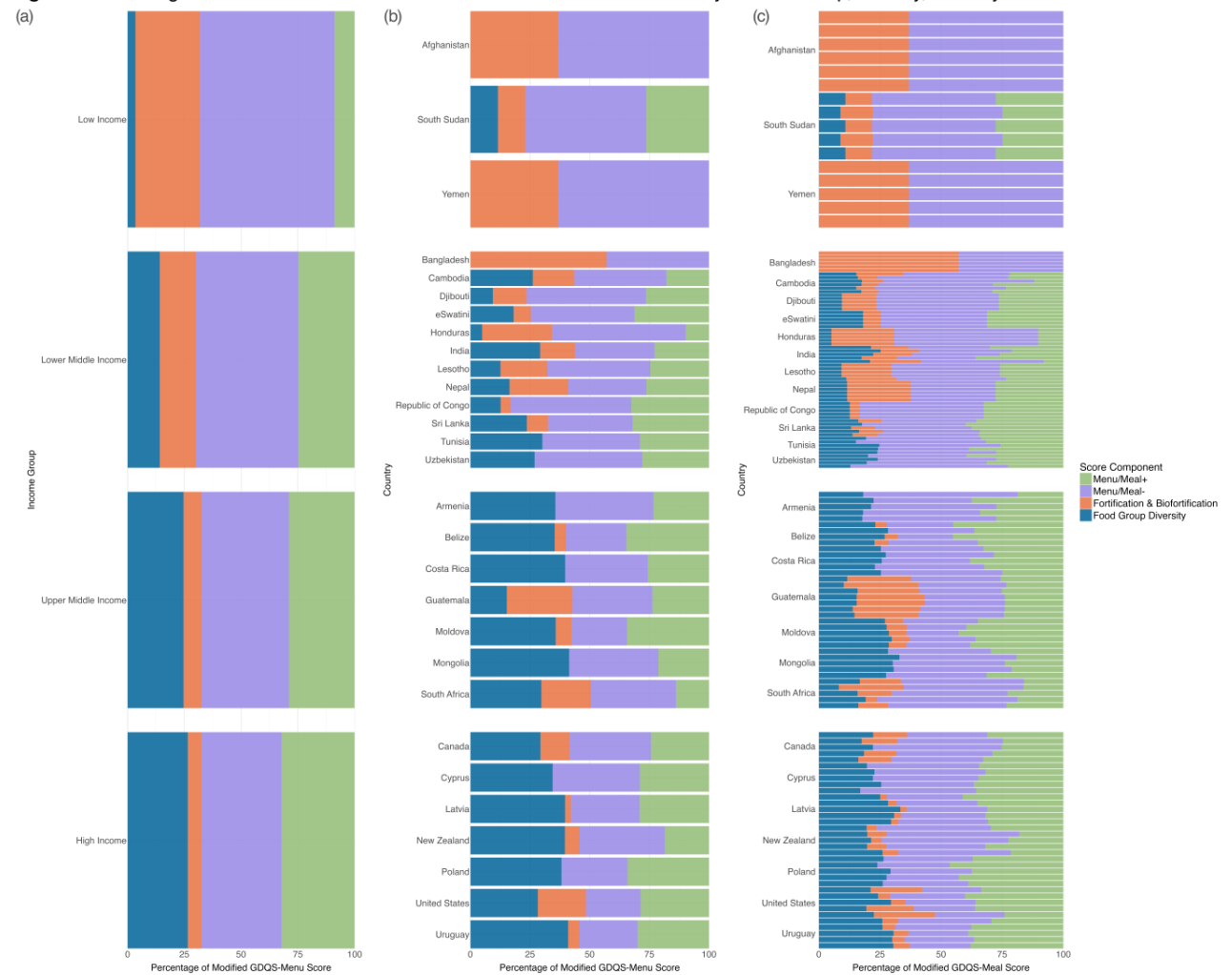
Figure 3: Modified GDQS-Menu Scores by Log-Transformed GDP per Capita.

Figure 4: Percentage Contribution of Modified GDQS-Menu and -Meal Sub-Metrics by Income Group, Country, and Day.*

*Note: Multiple rows per country in Panel C reflect different meal days